Non-equilibrium dynamics of polaron formation and polaron-polaron interaction in NiO

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In polarizable matter charge localization competes with delocalized electronic states and polaron formation is a dynamic process which occurs upon local lattice distortion [1]. The dynamics of corresponding electronic excitations is expected to differ fundamentally due to the corresponding relaxation phase space for Bloch electrons and polarons. The latter are candidates for metastable excitations with potential for opto-electronic applications [2].

In this talk recent results of time-resolved x-ray absorption [3] and optical spectroscopy [4] will be presented in order to investigate the formation dynamics and the interaction of polarons among each other in NiO. Many solid materials with strong electronic correlations exhibit spin fluctuations or magnetic order. Therefore, the spin degree of freedom can be expected to modify the nature of the polaron considerably [5,6].

Funding by the Deutsche Forschungsgemeinschaft through SFB 1242 and FOR 5249 – QUAST is gratefully acknowledged.

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